

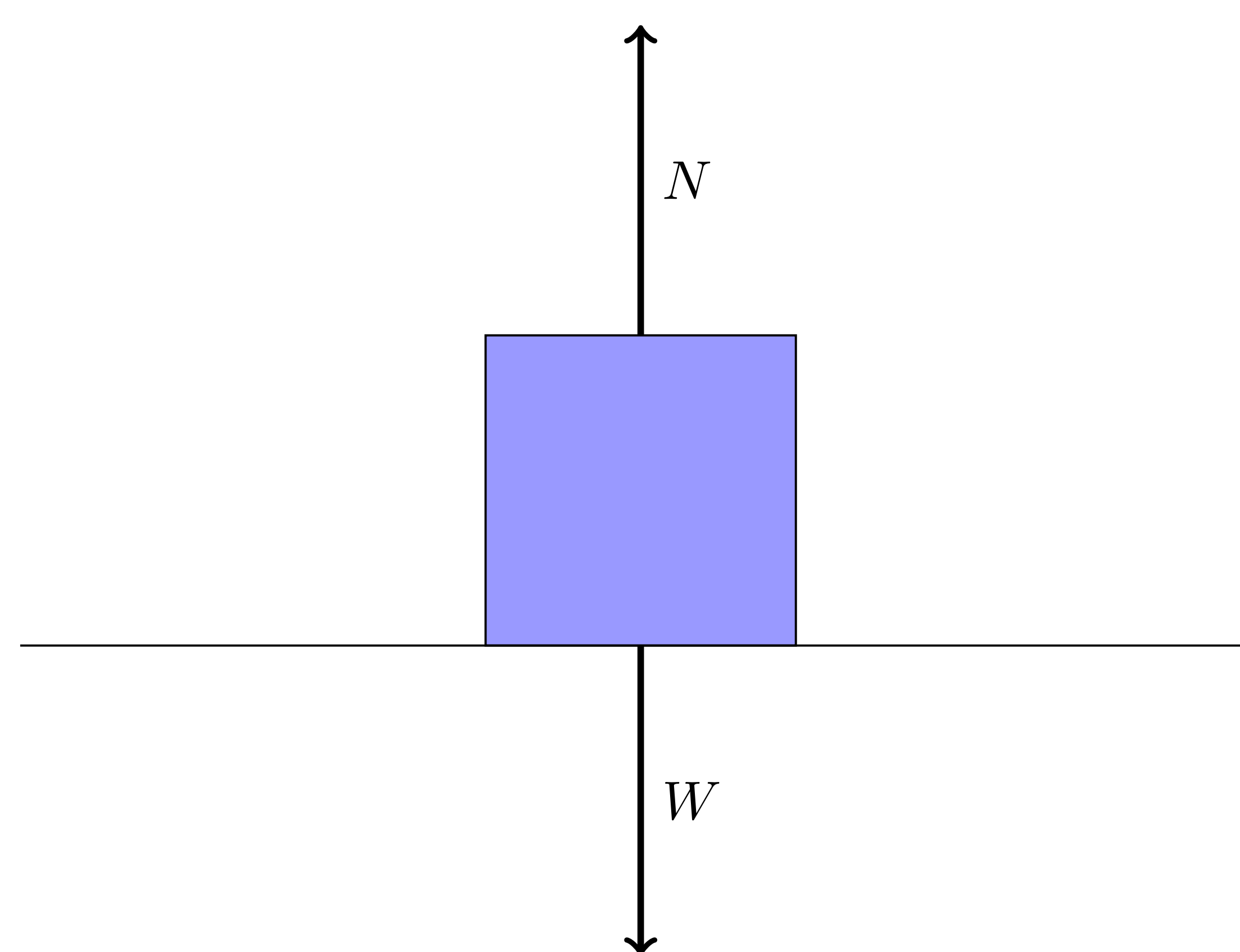
Group Problem Set 11: Forces and Applications of Newton's Laws: Free Body Diagrams and Breaking Up Forces

Physicists use the same approximate method for solving force problems that they use when solving kinematic problems. There are a few additional things that can be done to facilitate a better experience solving force problems:

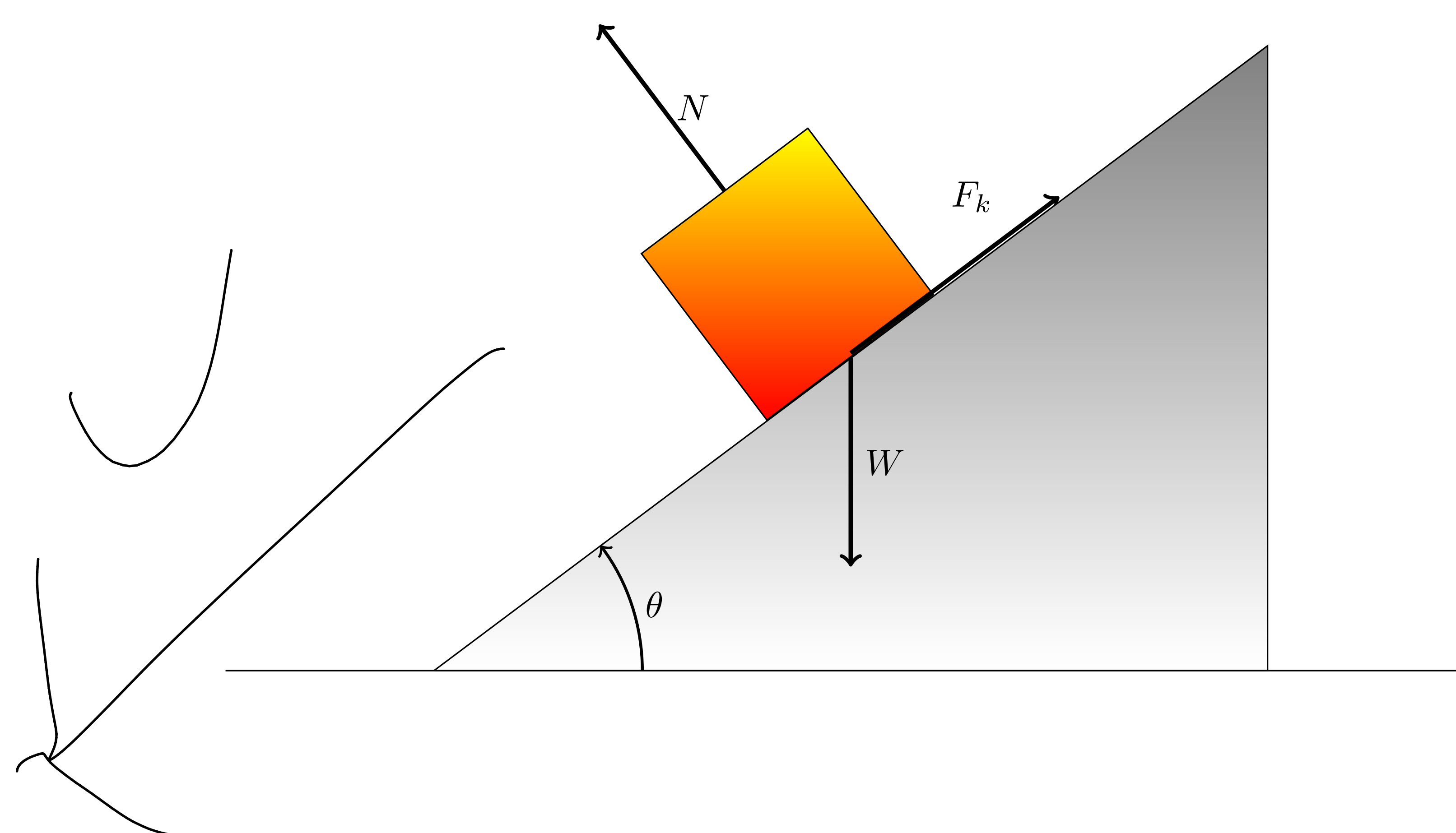
1. Draw a diagram of the problem at hand. This can be very simple (a box or a dot will do to represent the object of interest). If you have an idea of what the motion should be, go ahead and assign a coordinate system here.
2. Add the applied vectors to the object, with the vector labeled and pointing in the appropriate direction. (i.e. $\vec{W}$ should be pointing down toward where you have assigned the ground to be)
3. Determine which direction your object can move and assign an acceleration (direction doesn't matter for now). (i.e. a box cannot accelerate into a table, but can move along the table. The acceleration should be in the direction along the table).
4. If you haven't already done it, assign a coordinate system. Keep in mind that you will be considering motion due to acceleration, so choosing a coordinate system with one of the axes parallel to the direction of acceleration will make your life easier..
5. If any of your forces are applied at an angle to the coordinate system, write in that angle.
6. Construct a table listing all of the forces in each direction. Be sure to include the appropriate sign, (+) or (-), for each.
7. Sum the forces in each direction, applying Newton's 2nd law.
8. Solve
9. Check your units (force is measured in *Newtons* = $\frac{kg*m}{s^2}$!), does your magnitude make sense?

There are a couple of different force diagrams that you should familiarize yourself with. You will likely see or need to draw these diagrams for force related problems. The diagrams can and probably will be combined in problems (and in real life), so if you don't see a diagram here, you should at least have the fundamentals for what you need to build what is required.

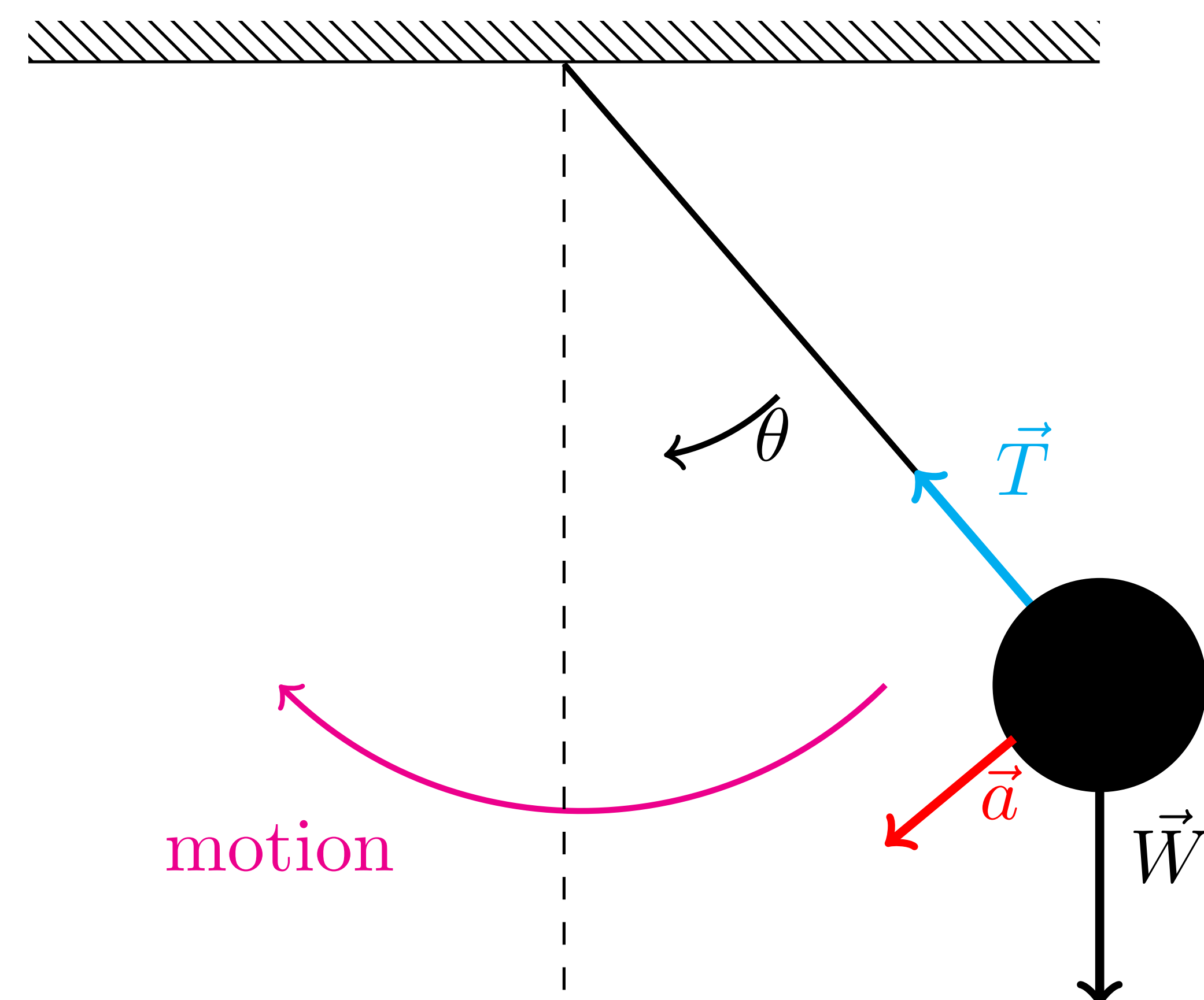
Mass on a Surface: Unless it is moving, a mass on a surface has a pretty straight forward summation of forces. If there is motion, it can only be to the right or left (perpendicular to the normal force).



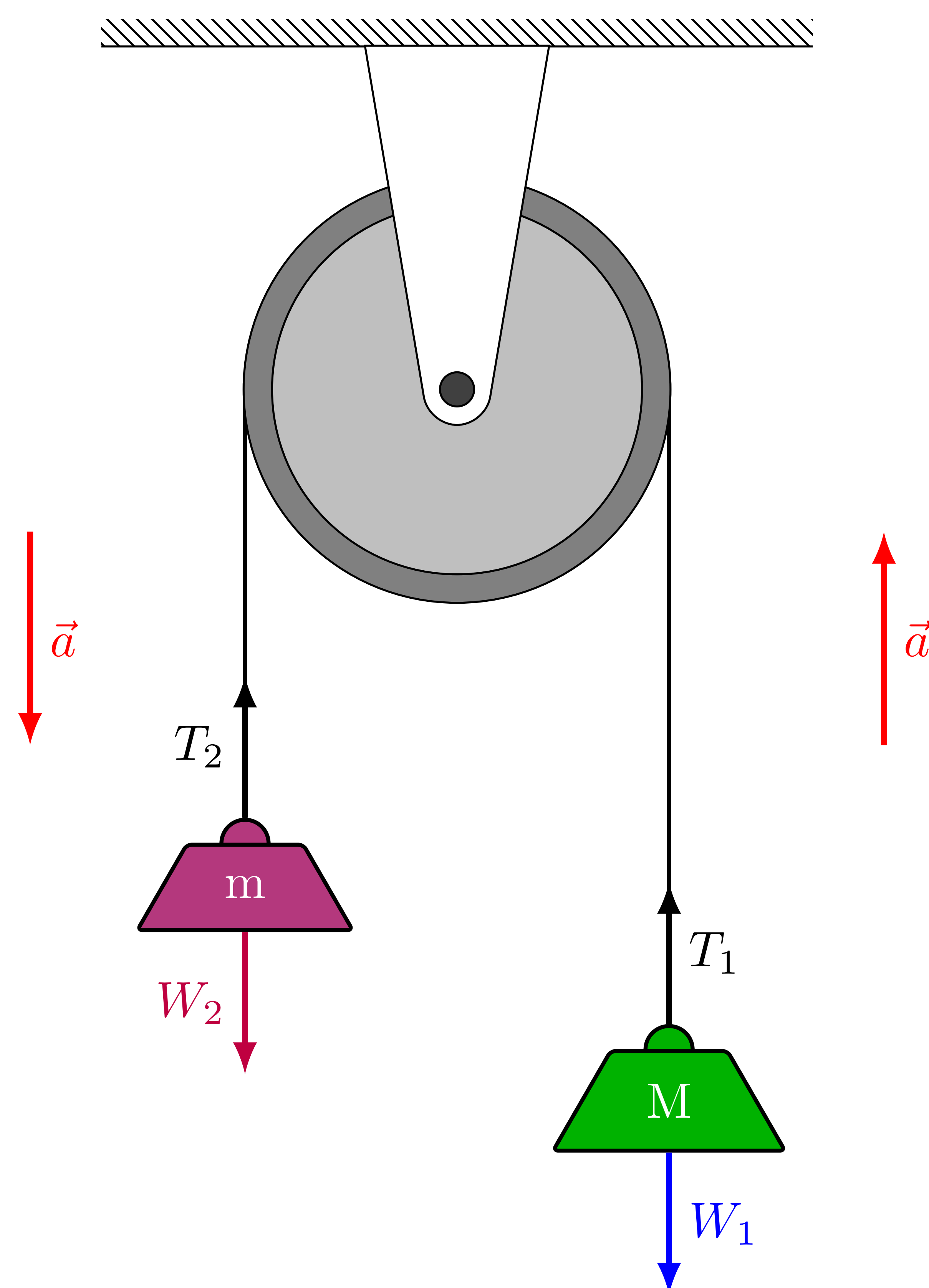
Mass on an Incline: Like the mass on the surface, the object of interest can only move left or right of where its starting position is. When the mass is on an incline, the complication is that the $\vec{F}_k$, $\vec{N}$ and $\vec{W}$ forces are now broken up into components specific to the incline's incident angle. The force due to static friction, $\vec{F}_k$, should point opposite the direction of motion. In the case where a block is at rest, it should be equal and opposite of the force due to gravity in the same component direction.



Pendulum: We've seen a pendulum, but we haven't applied forces. Like the mass on an incline, the pendulum will have a force due to gravity that has an angular dependence. Instead of a normal force, the pendulum has a force due to tension ($\vec{T}$). The motion of the pendulum will also have angular dependence.



Atwood Machines: Atwood machines consist of masses attached to a pulley. These machines are set up so that motion of one mass is **opposite** of the motion of the second mass. The pulley is generally treated as frictionless (but this changes later in the semester) and has no inertia (this will also change). The tether attached to each mass is generally taken to have negligible mass (for now).

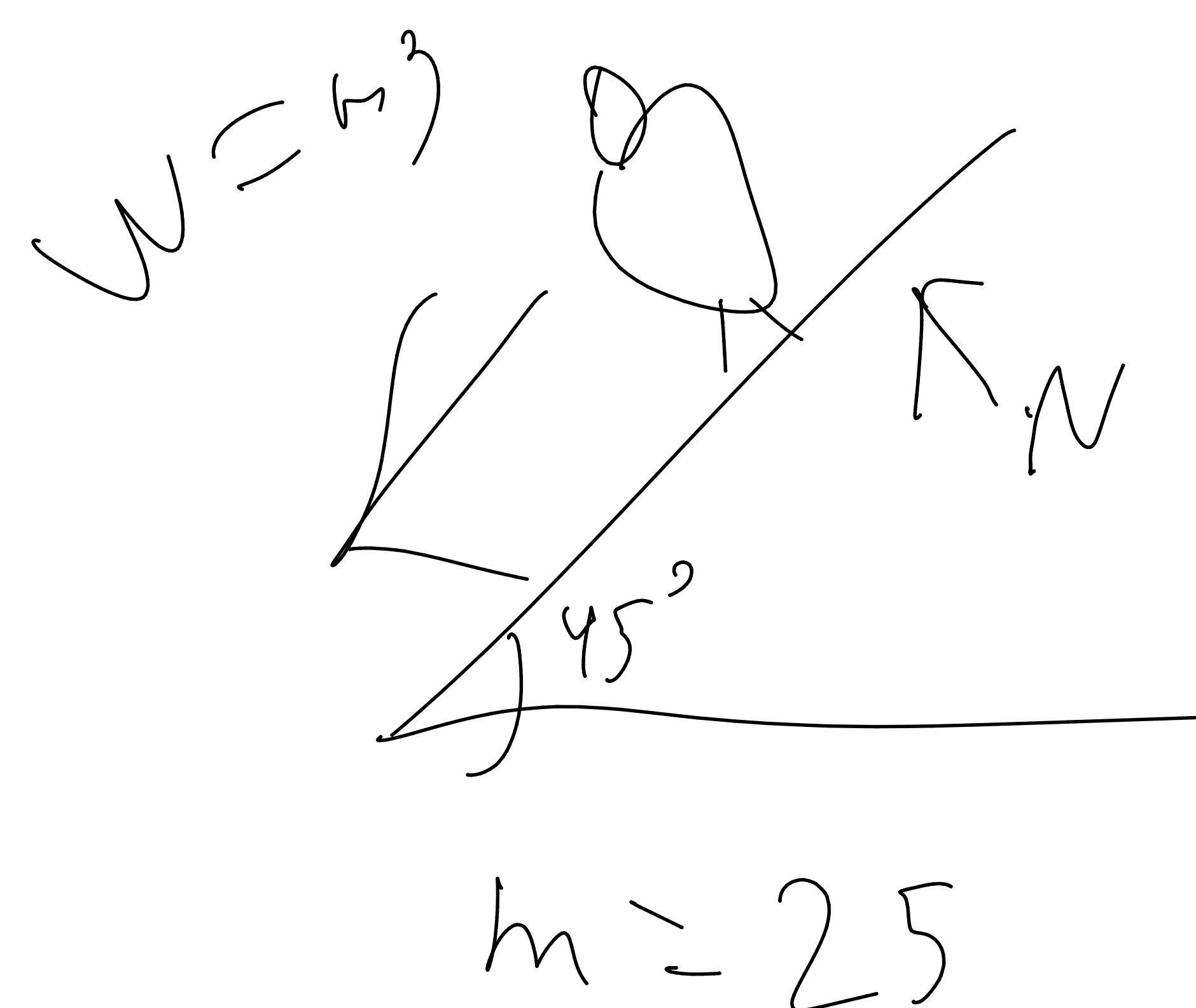


1. A 25 kg child slides down a steep slide on a play ground and experiences no friction. The slide sits at an angle of $\theta = 45^\circ$. Draw a free body diagram and complete the sum of forces table and answer the following questions.

Hint: What direction will the child slide? What is the direction of motion? Include these in your diagram. Choose a coordinate system such that the fewest forces are broken up into more than one component. Include this in your free body diagram. Using this coordinate system, fill out the table of forces and their $\hat{x}$ and $\hat{y}$ components on the right.

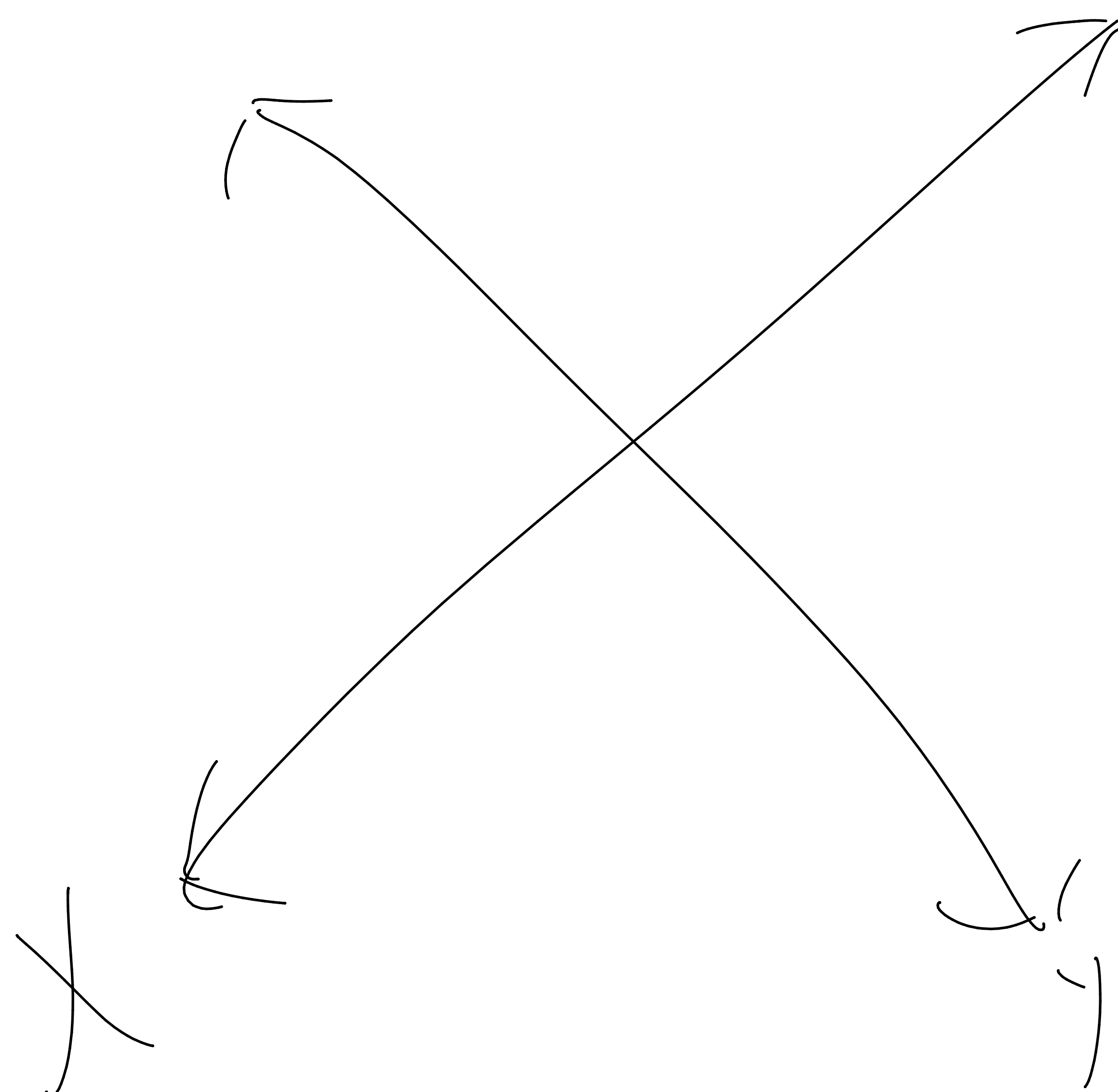
- (a) What is the acceleration of the child as they go down the slide?
- (b) Now imagine the child sits down at the top of the slide but doesn't move because friction is holding them in place. What is the force of friction keeping the child at the top of the slide? (if it helps, you can redraw the free body diagram and add this new force and then build a table to sum the forces).

Free Body Diagram



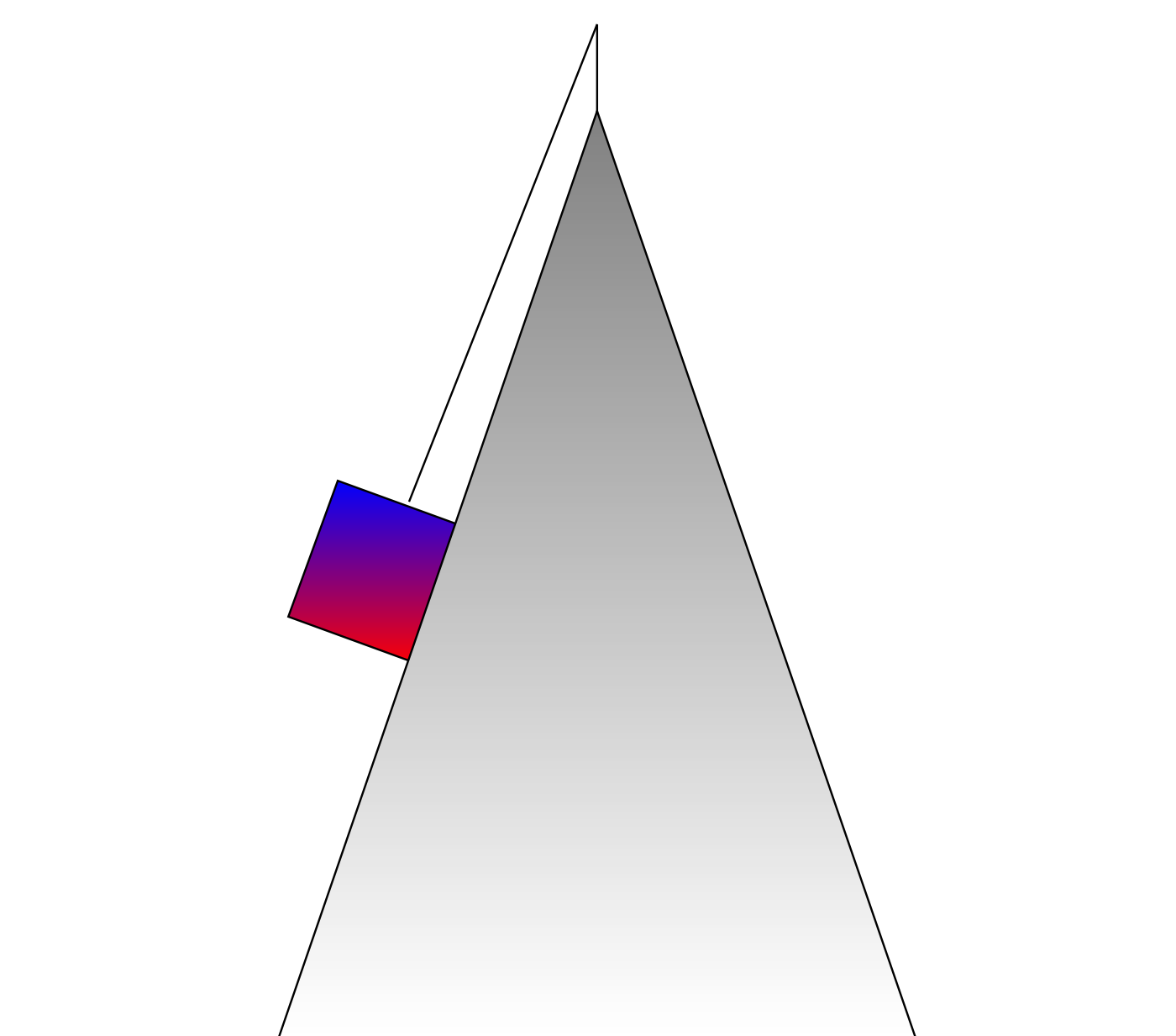
Hand-drawn equation: $g = 10 \text{ m/s}^2$

Force	$\hat{x}$ -component	$\hat{y}$ -component
N		
W		



2. Aaron (read A-A-RON) goes ice climbing in the rockies in the middle of January. Aaron finds himself tethered to an ice wall with a 70° incline and NO purchase (no friction)!!! Aaron has a mass of 100 kg.
- (a) What is the force due to tension that Aaron experiences while tethered to the mountain side?
- (b) Aaron's tether breaks and he free falls down the mountain with no friction. What Aaron's acceleration as he slides down the mountain side?

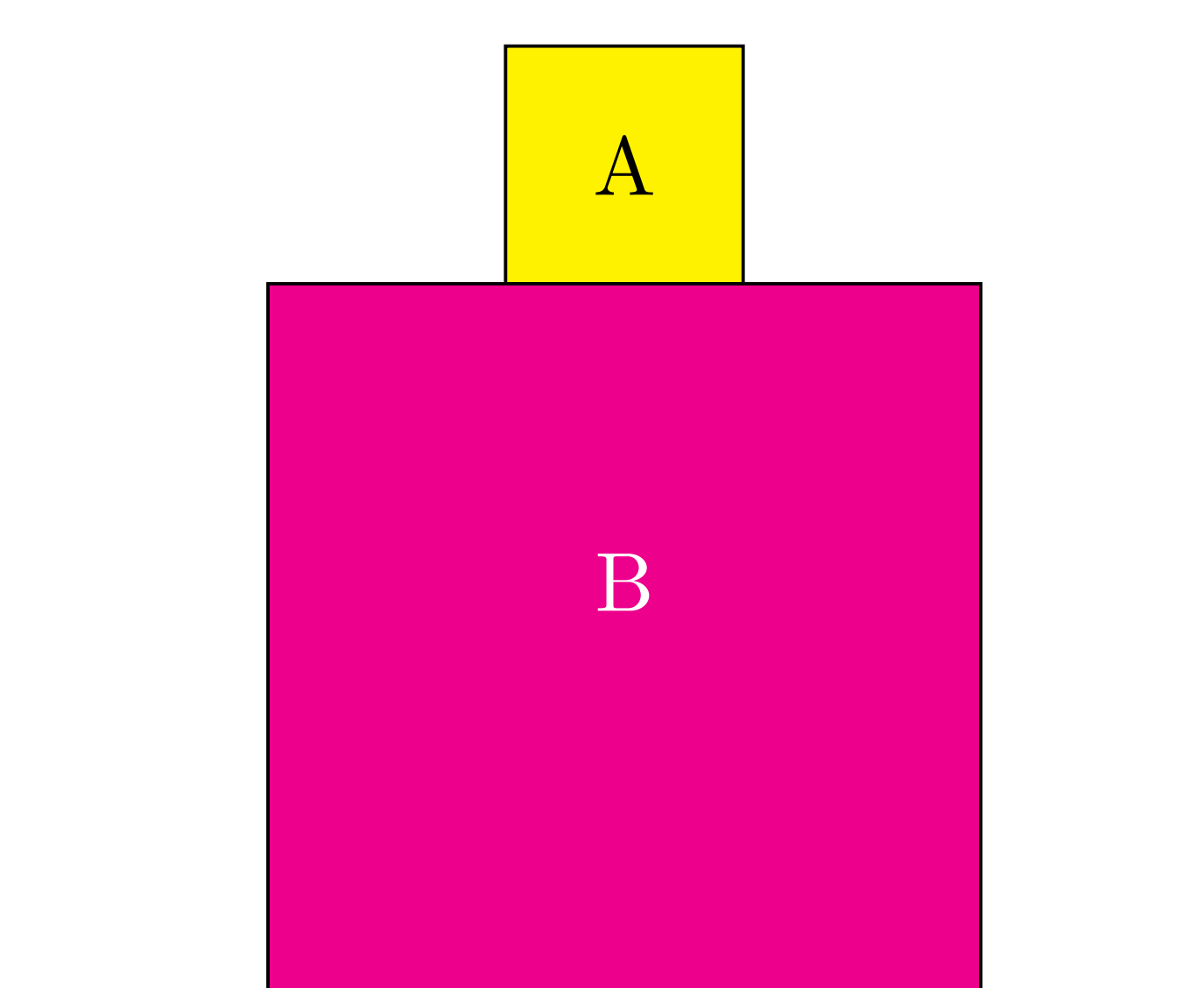
Free Body Diagram



Force	$\hat{x}$ -component	$\hat{y}$ -component

3. Two boxes are stacked one on top of the other. Draw the forces associated with Box A and Box B and write out the components in the table provided.

Free Body Diagram



Force	$\hat{x}$ -component	$\hat{y}$ -component

- Which box experiences a larger normal force?
- Which box exerts more force on the surroundings?
- Treating boxes A and B as one object, write the sum of forces below. Is the sum of forces table any different than when they are treated separately?
- Is the normal force on A+B (N_{A+B}) equal to the sum of individual normal forces ($N_A + N_B$)?
- What can you say about the sum of forces for objects considered individually and as a group?
- Box B is pushed along a surface such that the friction due to contact between boxes B and A, prevents Box A from falling. Draw the acceleration and frictional forces in the diagram and include this new force in the table provided. You can assume that Box B experiences no friction with the surface.
- Assume that the force applied causing the box to accelerate is $F = m_{A+B} * a$. Write the force required for box A to accelerate. What is the force required to move just box B at an acceleration a ?
- Does the friction required to keep box A on B depend on the mass of A, B or A+B? Why?